Chenghao (Cade) Li

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Summary

- Research Lead at Pika, driving fundamental research and end-to-end training of video generation models across pretraining, SFT, RLHF, distillation, and data; led post-training of the open-weights Pika H3 Flex (8.3× fewer denoising steps; top open-weights model on Design Arena), and currently lead post-training for Pika-LTX and an agentic omni video generation system.
- Foundation model training for Pika2.2 and Pika2.5 (14B); led PikaStream and Pikaformance, real-time and audio-driven video generation models.
- Previously at MiniMax on the Hailuo video (Video-01/02) and Image-01 foundation models, diffusion alignment, and 15B+ image data curation.

Experience

Pika 2025.03 – Present

Research Lead 2025.09 – Present

- **Post-training:** Led post-training of Pika H3 Flex, an open-weights omni video model built on MiniMax H3 through aesthetic-focused SFT, six-step distribution matching distillation (PDMD), and post-distillation reward learning (ReFL + DPO): 8.3× fewer denoising steps with improved visual and audio quality, ranked #7 overall and highest among open-weights models on Design Arena's text-to-video leaderboard at launch, and ∼1.8× faster denoising than Fal H3 Max at 768p. Supports text-, image-, and reference-conditioned generation and editing, single-step 4K upsampling, and five editable audio stems; weights and code are open (Hugging Face, GitHub). Also leading post-training of Pika-LTX across SFT, quality tuning (QT), RLHF, and distillation.
- **Agentic omni video generation:** Leading an agentic system on top of video models that takes any input (text, image, video, or references such as cast, voice, music, and footage), plans and routes each request, generates the soundtrack first and conforms the picture to it, and keeps character identity consistent across clips to deliver minute-scale videos from a single request.
- **Foundation model training:** Co-led training of the Pika2.5 14B foundation model.
- **Real-time model training:** Led PikaStream1.0, a real-time audio-driven video generation model for agents, across model design, pretraining, SFT, RLHF, and Self-Forcing distillation for real-time autoregressive generation; now leading PikaStream2.0 for real-time joint audio-video generation.
- **Data & systems:** Designed scalable data pipelines for video generation and lipsync training (filtering, MLLM reward models, human annotation); oversee production model APIs and video-/voice-agent systems, including Pika Speech.

Research Scientist 2025.03 – 2025.09

- **End-to-end foundation model training:** Led pretraining and SFT of the Pika2.2 10B foundation models for text-to-video, image-to-video, and keyframe-to-video generation.
- **Lipsync model training:** Led Pikaformance, Pika's lipsync model, through pretraining, DMD distillation, and post-training (SFT/RLHF), and shipped it in the Pika Social App with near real-time latency.
- **Video VAE:** Designed the Pika VAE with 16×16×4 spatiotemporal compression, combining convolutional and transformer encoder-decoders for efficient, high-fidelity video generation.

MiniMax (Hailuo AI) 2024.05 – 2025.03

Multimodal Algorithm Engineer

- Worked on the MiniMax-Video-01/02 (Hailuo) and MiniMax-Image-01 foundation models, across pretraining and large-scale data cleaning for text-to-image and text-to-video generation.
- Developed and deployed diffusion alignment (DPO, KPO) for text-to-image and subject-reference generation.
- Fine-tuned FLUX-based subject-reference modules and built decoupled personalization datasets to eliminate copy-paste artifacts.
- Built MLLM captioning, classification, and aesthetic models to curate a 15B+ image training set, and deployed a ComfyUI-based inference and evaluation service.

Earlier Experience

MiniMax – *3D Vision & AIGC Research Intern* 2021.11 – 2023.06

- Built NeRF-based animatable 3D avatars driven by audio, facial keypoints, SMPL, and pose, and set up a LightStage capture pipeline covering 500+ subjects.
- Fine-tuned Stable Diffusion; reimplemented Imagen, ControlNet, and T2I-Adapter; built controllable text-to-image features (style, pose, CFG, sampling acceleration) with FID / CLIP-score evaluation services.

Qualcomm – *AI Intern* 2019.07 – 2020.07

Neural network quantization and efficient inference for edge devices (data-free quantization, MobileNet, ShuffleNet).

Oceanbotech – *Robotics & Vision Intern* 2016.10 – 2019.06

ROS-based autonomous mobile robots with VSLAM, detection, and tracking on Jetson; pose control for an underwater ROV.

Selected Publications

Previous publications often appear under the name Shenghao Li.

- **S. Li**, Z. Xia, and Q. Zhao, "Representing Boundary-Ambiguous Scenes Online with Scale-Encoded Cascaded Grids and Radiance Field Deblurring," *IEEE TCSVT*, vol. 34, no. 4, 2024.
- **S. Li**, Q. Zhao, and Z. Xia, "Sparse-to-Local-Dense Matching for Geometry-Guided Correspondence Estimation," *IEEE TIP*, vol. 32, 2023.
- **S. Li**, S. Liu, Q. Zhao, and Q. Xia, "Quantized Self-Supervised Local Feature for Real-Time Robot Indirect VSLAM," *IEEE/ASME TMech*, vol. 27, no. 3, 2022.

Full list on Google Scholar.

Education

Shanghai Jiao Tong University – Ph.D., Pattern Recognition & Intelligent Systems 2020.09 – 2023.12

WeiChai Power Scholarship (top 1%, highest honor at SJTU) Advisor: Prof. Qunfei Zhao

East China University of Science and Technology – M.S., Mechanical Engineering 2017.09 – 2020.06

East China University of Science and Technology – B.Eng., Mech. Eng. & English 2013.09 – 2017.06

Skills

Methods: video & image diffusion models, VAEs, large-scale pretraining, SFT, RLHF / DPO / ReFL, few-step distillation (DMD, PDMD, Self-Forcing, rCM, AnyFlow), MLLM captioning & reward models, NeRF / 3D vision, SLAM

Tools: Python, C/C++, PyTorch, Transformers, ComfyUI, Linux **Languages:** English (IELTS 7.5), Chinese